

User Interface: Inspection, Monitoring, and Visualization

Overview

The orchestrator provides a dashboard UI for the user to:

- Monitor live runs -- watch key variables update each quarter
- Inspect completed runs -- drill into any quarter, any firm, any decision
- Compare across runs -- see how different regimes or seeds affect outcomes
- Aggregate all runs -- see patterns across the full simulation database

The UI is a local web application (Python + Streamlit or Panel) that reads from the run outputs and the cross-run database. It does NOT require internet access.

Dashboard Sections

1. Run Monitor (Live)

Shows the current run's progress in real time (updated after each quarter):

RUN: run_042 | Quarter: Q3 2034 (14/80) | Regime: baseline / baseline_gaap
Elapsed: 45 min | Est. remaining: 2h 30min

INDUSTRY OVERVIEW	
Active firms: 5	Total patients: 12,400
Total revenue: \$1.2B/quarter	HHI: 2,150
Avg price: \$72,000	Max generation: Gen 2
FIRM PERFORMANCE (sparklines)	
Firm 0 (Aeterna): Revenue	■ \$280M ↑12%
Firm 1 (GenVita): Revenue	■ \$245M ↑6%
Firm 2 (NovaLife): Revenue	■ \$120M ↓15%
Firm 3 (BioAge): Revenue	■ \$310M ↑20%
Firm 4 (Senova): Revenue	\$190M ↑2%
FINANCIAL HEALTH	
Cash runway (quarters): 8, 5, 3(!), 12, 7	
Debt/equity: 0.3, 0.8, 2.1(!), 0.2, 0.5	
Bank capital: CBank 92% CFund 88%	
LATEST GAZETTE HEADLINE	
"BioAge Captures Lead with Gen 2 Launch; NovaLife Cash Position Raises Concerns"	

2. Time-Series Plots (Single Run)

Interactive charts for any completed (or in-progress) run:

Revenue and Market Share

- Line chart: each firm's quarterly revenue over time
- Stacked area: market share evolution
- Event markers: defaults, entries, M&A, clinical holds

Profitability

- Line chart: gross margin, operating margin, net margin per firm
- Bar chart: net income per firm per quarter (positive/negative)
- Cumulative profit/loss by firm

Balance Sheet

- Stacked bar: asset composition (cash, AR, inventory, PPE, intangibles, goodwill)
- Stacked bar: liability + equity composition
- Line chart: debt/equity ratio over time

Cash Flow

- Waterfall chart: CFO, CFI, CFF, net change per firm per quarter
- Line chart: cumulative cash position
- Cash runway estimate over time

R&D and Technology

- Line chart: R&D spending per firm over time
- Step chart: product generation achieved per firm
- Bar chart: R&D allocation (product/process/delivery) per firm

Market Dynamics

- Line chart: average price, high price, low price over time
- Line chart: total demand (patients) over time
- Scatter: price vs. market share per firm per quarter

Equity Valuation

- Line chart: stock price per firm over time
- Line chart: pricing error ($P - P^*$) over time
- Bar chart: IPO valuations for all entrants

Financial Institutions

- Line chart: bank capital over time
- Bar chart: cumulative interest income vs. losses
- Pie chart: loan portfolio by firm

Scoring

- Table: equity IRR, multiple, key metrics per firm-incarnation
- Radar chart: environment ratings by dimension
- Bar chart: institution loss rates

3. Single-Quarter Drill-Down

Click on any quarter to see:

Market outcomes:

- Who sold how much, at what price, at what market share

- Events that occurred
- Environment narrative (the gazette)
- Demand model baseline vs. actual outcome

Per-firm details:

- Full income statement, balance sheet, cash flow statement
- Raw decision (what the firm requested)
- Clamped decision (what actually happened)
- Reasoning trace (the firm's multi-step reasoning)
- Dossier snapshot

Financial institution details:

- Terms offered to each firm
- Reasoning for each term
- Portfolio state

Orchestrator actions:

- Feasibility clamping details (what was clamped and by how much)
- Settlement details (revolver draws, equity issuance)
- Validation results (which invariants checked)
- Warnings/flags

4. Cross-Run Comparison

Select 2+ runs to compare:

Side-by-side line charts:

- Same metric, different runs (e.g., "revenue for run_042 vs. run_043")
- Overlay with confidence bands if multiple seeds available

Regime comparison:

- Table: key outcome metrics by regime (rows = regimes, columns = metrics)
- Box plots: distribution of outcomes across seeds for each regime

Scatter plots:

- X = one strategy metric (e.g., avg R&D spend), Y = outcome (e.g., equity IRR)
- Colored by regime or fingerprint style
- Across all firm-incarnations in the database

5. All-Runs Aggregate Dashboard

Summary across the entire simulation database:

Database statistics:

- Total runs: N
- Total firm-incarnations: M
- Total quarters simulated: K
- Default rate: X%
- Average equity IRR: Y%

Strategy-outcome analysis:

- Heatmap: fingerprint_style x outcome_metric

- "Which personality types perform best under which regimes?"

Environment quality tracking:

- Line chart: average environment rating over successive runs
- "Is the environment agent improving with cross-run learning?"

Regime comparison (aggregated):

- Table: all regimes tested, with mean and std of key outcomes
- Statistical test results (if enough runs): "Does R&D capitalization significantly change R&D spending?"

Implementation

Technology Choice

Streamlit is recommended:

- Pure Python (no JavaScript needed)
- Interactive widgets (sliders, dropdowns, date pickers)
- Native support for pandas DataFrames and matplotlib/plotly charts
- Runs locally (streamlit run dashboard.py)
- Single file for simple dashboards, modular for complex ones

File Structure

```
llm_firm_lab/
  dashboard/
    app.py                # Main Streamlit app
    pages/
      01_run_monitor.py   # Live run monitoring
      02_time_series.py    # Single-run time-series plots
      03_quarter_drill.py  # Single-quarter detail view
      04_cross_run.py      # Cross-run comparison
      05_aggregate.py      # All-runs aggregate analysis
    components/
      charts.py            # Reusable chart components
      data_loader.py        # Load Compustat, debrief, scores, dossiers
      formatters.py         # Statement formatting for display
```

Data Sources

The dashboard reads from:

- data/compustat_q.csv (all runs)
- data/debrief.csv (all runs)
- data/scores.csv (all runs)
- data/run_summaries.csv (all runs)
- outputs/{run_id}/ (per-run details: statements, dossiers, gazettes, reasoning)

Running the Dashboard

```
# View a specific run
```

```
python -m llm_firm_lab dashboard --run-id run_042

# View all runs
python -m llm_firm_lab dashboard

# Monitor a live run
python -m llm_firm_lab dashboard --live --run-id run_042
```

Plot Export

All plots can be exported to:

- PNG/SVG (for papers)
- Interactive HTML (for sharing)
- PDF (for reports)

CLI Inspection (Non-GUI Alternative)

For quick checks without launching the dashboard:

```
# Summary of a run
python -m llm_firm_lab inspect --run-id run_042

# Print a firm's income statement for a specific quarter
python -m llm_firm_lab inspect --run-id run_042 --firm firm_0 --quarter 14 --statement is

# Print the gazette for a quarter
python -m llm_firm_lab inspect --run-id run_042 --quarter 14 --gazette

# Print a firm's reasoning for a quarter
python -m llm_firm_lab inspect --run-id run_042 --firm firm_0 --quarter 14 --reasoning

# Quick aggregate statistics
python -m llm_firm_lab inspect --all-runs --summary

# Export plots to PNG
python -m llm_firm_lab plot --run-id run_042 --type revenue --output plots/
python -m llm_firm_lab plot --all-runs --type strategy_outcomes --output plots/
```

Configuration

```
dashboard:
  port: 8501
  auto_refresh_seconds: 30      # for live monitoring
  default_chart_library: "plotly" # plotly | matplotlib
  export_format: "png"
  theme: "light"                # light | dark
```